ARJUN JOSHI

AI Systems Engineer | Simulation Architect | Edge Computing Specialist
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Professional Summary

AI Systems Engineer & Systems Integrator specializing in hardware-in-the-loop (HIL) simulation, edge computer vision, and end-to-end product design. Architect of twin-environment platforms where high-fidelity driver simulation, physical rigs, and edge devices share unified telemetry and safety indices. Design multi-stage ML/AI pipelines (statistical models → GBM/RL → LLM policy layers) and optimize lightweight CV models for low-cost edge hardware. Draw on a quantitative background in biostatistics and stochastic modeling to validate AI behavior in noisy, real-world environments.

Technical Skills

Simulation & Physics: CARLA Simulator, Hardware-in-the-Loop (HIL), Unreal Engine Interop, Moza Racing Hardware, Arduino I/O.

Edge AI & Hardware: NVIDIA Jetson (Orin Nano), TensorRT, DeepStream, Edge Impulse, GPIO/Sensor Integration.

ML, AI & Computer Vision: scikit-learn, PyTorch, NumPy, Pandas, Matplotlib, YOLOv8/v11, MediaPipe, LLM Integration (Llama), Reinforcement Learning (RL).

Data, Infrastructure, Environments: Python, C++, Docker, Linux, Bash Scripting, PyArrow, Apache Parquet, Asyncio, SQL, Jupyter

Statistical Analysis: Bayesian Inference, Stochastic Modeling, Time-Series Analysis (VAR, LSTM), Experimental Design.

PROFESSIONAL EXPERIENCE

QRyde / HBSS Connect Corp | Lowell, MA
AI Simulation Engineer - Lead Architect & System Integrator April 2025 – Present

Lead architect and end-to-end systems integrator leading a team of 2 engineers for the Q-DRIVE Safety Ecosystem, a closed-loop platform that connects high-fidelity driver simulation, physical driver rigs, edge computer-vision devices, and real-world vehicle telemetry under a unified data schema. Joined corporate team as CV specialist in November 2025.

- **Progression & Rapid Deployment**: Fast-tracked from Engineering Intern to AI Simulation Engineer in 6 weeks and given ownership of Q-DRIVE and a two-person engineering team after the second national showcase (APTA Boston), driving the product ecosystem from concept to deployment with MART Transit Authority as alpha partner.
- **System Integration & Ownership**: Designed and built a production-grade hardware-in-the-loop driver safety simulator using a high-fidelity real-time vehicle dynamics engine with custom physics, custom control electronics/switch panels, a direct-drive force-feedback steering system, and a dedicated cockpit to capture realistic driver behavior and vehicle dynamics.
- **Predictive Stack Design**: Structured the twin simulation/edge environment around a statistical safety model and unified feature schema, providing a foundation for upcoming gradient-boosted and reinforcement-learning policies, with an LLM-driven policy layer and PyArrow/Parquet backend already in place for natural-language control and analytics.
- **Edge AI Optimization**: Designed a retrofit ADAS + DMS edge stack for public transit using fine-tuned lightweight vision models on NVIDIA Jetson Orin Nano, leveraging proxy features instead of heavy segmentation networks. Applied TensorRT optimization to raise on-device throughput from ~28 FPS to 60+ FPS, enabling a multi-camera safety system at well under 50% initial cost point of typical 6-point ADAS/DMS solutions.
- **LLM-Orchestrated CV Solutions**: Designing edge-deployed CNN/VLM-based tracking systems for corporate clients, integrated with a centralized AI policy/workforce platform that turns natural-language policies into validated events and actionable monitoring rules, while partnering with MART Transit Authority to validate human-in-the-loop data collection and deliver integrated demos at CTAA, APTA, and CTA national summits.

J & J Asset Management | Boston, MA & Washington DC Metro & Tampa
Senior Development Asset Manager 2019-2024

- **Predictive Analytics:** Led data-driven investment strategies using Python predictive modeling and Tableau dashboards to identify high-yield opportunities, informing **$2M+ in property acquisitions**.
- **Strategic Operations:** Navigated high-friction regulatory environments, successfully managing rezoning litigation and neutralizing aggressive competitive threats. Orchestrated complex workflows between contractors, legal teams, and tenants to secure asset viability.

- **Impact Metric Development:** Engineered a weighted-sum ecological impact model for property development, quantifying artificial ecological pressures to guide sustainable investment decisions.

NOAA (National Oceanic and Atmospheric Administration) | San Diego, CA
Contracted through Ocean Associates, Inc.
Biostatistical Modeler 2016 – 2018

- **Stochastic Modeling:** Developed predictive models for biological resource variability in **high-uncertainty environments**. Applied **Bayesian inference** to datasets characterized by high noise and non-deterministic behavior—establishing the rigorous statistical methodology now applied to ML validation.
- **Data Integrity Pipeline**: Managed the ingestion and validation of large-scale biological datasets for federal regulatory reporting. Maintained data integrity for a 500K+ record Oracle SQL database, ensuring 100% compliance with strict data governance standards.

EDUCATION

Marine Science & Technology (M.S. Program) || University of Massachusetts, SMAST
Focus: Quantitative modeling, complex systems, biostatistics, experimental design.

B.S.: Marine Science, *with Distinction* || Boston University - Marine Program (BUMP)
Focus: Quantitative ecology, population dynamics, field-to-lab research pipelines; 2× UROP grant recipient and undergraduate research team lead.

Diploma: Data Science & Machine Learning || BrainStation
Focus: Applied machine learning, deep learning, NLP, and data science engineering (Python, SQL, EDA, model evaluation; AWS, Hadoop, Spark).

- *Capstone*: Anthropogenic Impact Modeling – end-to-end ETL and predictive pipeline for ecosystem health using LSTMs and Vector Autoregression (VAR).

PROJECTS & OPEN SOURCE

Q-DRIVE Safety Simulator (GitHub: `Carla_AI_dev`, `orin_jps_scripts`)
Hardware-in-the-loop driver safety simulator with force-feedback steering, microcontroller I/O, multi-monitor Unreal rendering, and a twin-environment PyArrow/Parquet telemetry pipeline feeding predictive safety indices and an LLM/STT scenario-orchestration layer.

Anthropogenic Impact Index (GitHub: `Tampa_Apollo_AnthropogenicIndex`)
Capstone project modeling ecosystem health from urbanization and water-quality data via a Python/Pandas/SQL ETL pipeline and VAR + LSTM forecasting to produce a composite ecosystem stress index.

Vision Assist – OCR/VLM Agent (GitHub: `donkey/vision_assist_cl`)
Screen-aware vision assistant that combines live OCR with vision-language models (VLMs such as Qwen-VL) to interpret on-screen content, validate events, and drive agentic workflows from natural-language instructions.